

ISIT, Russian Federation

WMO: 249510

Lat: 60.817N

Lon: 125.317E

Elev: 428

StdP: 14.47

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-51.4	-48.3	-55.5	0.2	-51.4	-52.6	0.2	-48.3	19.2	-20.8	18.1	-21.5	1.5	250	0.671

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	83.1	67.6	79.5	65.8	75.9	64.0	69.8	79.4	67.6	76.6	65.5	73.7	4.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	98.5	75.2	64.1	91.1	72.6	62.1	84.6	70.6	34.2	79.5	32.4	77.0	30.7	73.6	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.7	14.5	12.4	DB	-54.0	90.4	4.1	3.1	-57.0	92.6	-59.4	94.4	-61.7	96.1	-64.7	98.4	(4)
			WB	-54.0	73.4	4.1	2.1	-57.0	74.9	-59.4	76.1	-61.7	77.3	-64.7	78.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.0	-27.9	-20.8	-0.7	25.2	44.6	59.7	65.1	59.0	44.0	23.3	-7.7	-26.5	(6)
	DBStd	35.54	14.35	13.24	14.42	10.54	7.37	7.18	6.50	6.59	7.06	10.71	15.83	13.59	(7)
	HDD50	12064	2415	1982	1573	744	200	10	1	8	203	828	1730	2370	(8)
	HDD65	16554	2880	2402	2038	1194	632	185	78	207	630	1293	2180	2835	(9)
	CDD50	1116	0	0	0	0	33	302	470	288	23	0	0	0	(10)
	CDD65	131	0	0	0	0	0	28	82	21	0	0	0	0	(11)
	CDH74	1165	0	0	0	0	4	272	703	184	2	0	0	0	(12)
	CDH80	271	0	0	0	0	0	51	189	31	0	0	0	0	(13)
	WSAvg	4.0	4.3	3.9	3.3	3.7	4.5	3.8	3.7	3.9	4.4	4.2	4.3	4.2	(14)
	PrecAvg	11.4	0.5	0.3	0.3	0.4	0.9	1.7	2.1	2.0	1.4	0.8	0.6	0.5	(15)
Precipitation	PrecMax	18.0	1.0	0.6	0.8	1.0	3.5	3.4	3.9	4.1	3.9	1.3	1.0	0.9	(16)
	PrecMin	7.3	0.1	0.1	0.1	0.0	0.1	0.3	0.2	0.1	0.6	0.2	0.2	0.2	(17)
	PrecStd	2.7	0.2	0.1	0.2	0.2	0.7	0.9	0.8	1.2	0.8	0.4	0.2	0.2	(18)
	PrecStd	2.7	0.2	0.1	0.2	0.2	0.7	0.9	0.8	1.2	0.8	0.4	0.2	0.2	(18)
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.3	20.5	40.4	55.7	73.4	85.2	90.1	83.6	68.5	48.2	34.7	17.8	(19)
		MCWB	15.2	18.8	32.5	43.2	55.3	65.2	71.0	68.5	56.2	40.7	32.3	16.8	(20)
	2%	DB	5.9	12.5	32.2	48.9	67.4	80.7	85.1	79.0	62.7	42.8	29.4	6.6	(21)
		MCWB	5.0	11.3	27.5	38.9	52.0	63.9	68.9	65.7	53.4	38.7	27.8	5.8	(22)
	5%	DB	0.0	5.1	27.2	44.1	62.7	77.0	81.3	75.0	58.6	39.2	21.8	-0.7	(23)
		MCWB	-0.7	4.2	23.8	36.0	49.5	62.5	67.5	64.2	51.3	35.7	19.9	-1.4	(24)
	10%	DB	-6.7	-1.3	21.8	40.4	58.4	73.0	77.7	71.2	55.2	36.3	14.8	-7.0	(25)
		MCWB	-7.2	-1.9	19.1	33.8	47.2	60.5	66.0	61.9	49.6	33.2	13.6	-7.5	(26)
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.5	19.0	32.9	43.5	56.8	69.1	73.5	70.8	58.0	42.7	32.6	17.0	(27)
		MCDB	16.7	20.2	39.3	55.0	70.4	80.8	85.0	80.5	65.0	45.6	34.2	18.0	(28)
	2%	WB	5.0	11.1	27.8	39.6	53.1	66.0	71.1	67.6	55.0	38.9	27.7	5.8	(29)
		MCDB	5.6	12.4	32.2	47.9	64.6	76.6	81.7	75.7	60.9	42.4	29.3	6.6	(30)
	5%	WB	-0.7	4.2	23.9	36.7	50.5	63.9	69.2	65.4	52.5	36.0	20.0	-1.3	(31)
		MCDB	-0.1	5.1	27.3	42.9	60.8	74.0	78.6	72.8	57.3	38.7	21.7	-0.6	(32)
	10%	WB	-7.2	-2.0	19.2	34.4	48.1	61.8	67.3	63.2	50.1	33.3	13.5	-7.6	(33)
		MCDB	-6.7	-1.4	21.8	39.8	56.9	70.9	76.0	69.9	54.6	35.9	14.9	-7.2	(34)
Mean Daily Temperature Range	5% DB	MDBR	11.2	14.4	22.0	22.1	18.6	20.3	19.3	18.2	14.5	10.8	11.1	10.2	(35)
		MCDBR	16.6	18.9	22.7	23.0	26.7	26.3	25.1	25.0	21.6	12.1	16.9	17.7	(36)
		MCWBR	15.9	17.8	18.9	14.8	14.7	12.7	12.5	14.1	13.8	9.4	16.3	17.3	(37)
	5% WB	MCDBR	16.3	18.8	22.3	20.3	24.4	22.7	21.5	21.2	19.0	11.4	16.5	17.4	(38)
		MCWBR	15.9	18.0	18.8	13.4	14.1	12.1	12.0	13.0	12.8	9.4	16.1	17.1	(39)
Clear-Sky Solar Irradiance	taub		0.239	0.274	0.308	0.365	0.427	0.427	0.516	0.464	0.351	0.295	0.240	0.210	(40)
	taud		1.965	2.039	2.052	2.038	2.125	2.241	1.976	2.089	2.392	2.305	2.002	1.892	(41)
	Ebn at Noon		149	220	253	259	251	254	224	226	242	216	147	105	(42)
	Edh at Noon		14	25	35	43	43	39	50	42	26	19	13	9	(43)
All-Sky Solar Radiation	RadAvg		144	469	1015	1613	1678	1829	1655	1268	784	400	193	76	(44)
	RadStd		10	21	37	56	109	137	129	83	77	47	17	4	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.44	N/A	N/A	N/A	N/A	N/A	-488	-500	N/A	N/A
Regional (3 neighbors)		+1.44	N/A	N/A	N/A	N/A	N/A	-488	-500	N/A	N/A

Nomenclature: See separate page